

1(a)(i)	2 ;	1
1(a)(ii)	pumps / actively transports, protons / hydrogen ions, into intermembrane space ; <i>ref. to</i> proton gradient ;	2
1(a)(iii)	<i>any three from:</i> 1 ATP synthase ; 2 protons / hydrogen ions, move (from intermembrane space) to matrix ; 3 by facilitated diffusion ; 4 ATP synthesised from ADP + Pi ; A ADP is phosphorylated to ATP 5 <i>ref. to</i> chemiosmosis ;	3

Question	Answer	Marks
1(b)	any four from: 1 electron transport chain / ETC, stops / reduced ; 2 oxygen does not act as (final) electron acceptor ; 3 no / less , protons pumped into intermembrane space or no / less steep, proton gradient ; 4 build-up of, reduced NAD / reduced FAD or no / less, reduced NAD / reduced FAD, oxidised / recycled or no / less, NAD / FAD, regenerate / recycled ; 5 no / less, ATP production or no / fewer, protons pass through ATP synthase ; 6 Krebs cycle / link reaction / oxidative phosphorylation, stops / reduced ; 7 <i>ref. to</i> glycolysis / anaerobic respiration / lactate produced / ethanol produced / substrate level phosphorylation ;	4